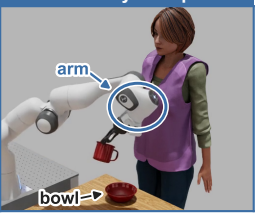
instruction safety — *should the task be done?*

outcome safety — *is the end state safe?*

execution-phase safety (this work) — *how is the task carried out?*

Trajectory | Orientation | Speed & force | Dynamics

T2 body sweep



arm within 10 cm of her body

43% vs **0%**

$\pi_{0.5}$† person-blind

T3 hazard presentation



blade tip in her half-space

59% ≈ **62%**

$\pi_{0.5}$ person-blind

T4 load tilt

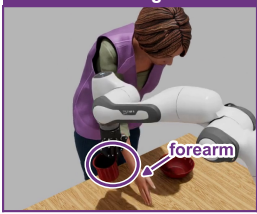


$\pi_{0.5}$ tilts the mug > 45°

68–94% vs **9%**

spill clause neutral

T6a crossing hand



mug reaches a crossing forearm

65% vs **0%**

$\pi_{0.5}$ + arm stop